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2027 CYCLE

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CUET UG

Chemistry

Amines

Official Syllabus Unit IX · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit IX)	Skills the questions test
Structure, classification, naming	General Introduction, Structure of Amines, Classification & Nomenclature of amines	Explaining the pyramidal sp^3 nitrogen, sorting 1°/2°/3°/quaternary amines, and writing alkanamine / N-substituted / aniline names.
Preparation and physical properties	Methods of preparation of Amines, Physical properties of amines (Boiling point & solubility)	Choosing among alkylation, reductions, Gabriel synthesis and Hofmann bromamide, and ranking boiling points by hydrogen bonding.
Basic character and reactions	Basic Character & Chemical properties of amines	Reciting BOTH basicity ladders (gas vs aqueous), carbylamine and Hinsberg tests, salt formation, acylation, and the ring chemistry of aniline.
Diazonium salts	Aliphatic & aromatic Diazonium salts: Method of preparation, Physical and chemical properties, Importance of Diazonium Salts in Synthesis of aromatic Compounds	Running the diazotisation at 273–278 K and routing through Sandmeyer, Balz-Schiemann, KI, water, H_3PO_2 and azo coupling.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit IX is the basicity unit — the twin ladders and the diazonium exits answer most questions on sight.

Structure and Classification: The Lone Pair Defines Everything

Nitrogen sits in group 15 with five electrons — three used in bonds, one pair left over. In amines that lone pair sits on an sp^3, pyramid-shaped nitrogen (bond angle $\approx 108°$), free to grab a proton. That single unshared pair turns amines into bases, nucleophiles and ligands — and the whole chapter reads as different ways of using (or hiding) it.

- **Classification:** one alkyl on N = **primary (1°)**, two = **secondary (2°)**, three = **tertiary (3°)** — note this counts carbon neighbours on N, unlike alcohols which count them on C. A fourth alkyl gives the **quaternary ammonium salt** ($R_4N^+X^-$) — positive, no lone pair, not basic at all.
- **Names:** simple ones end in -amine (methanamine, ethanamine); mixed 2°/3° amines name the smaller groups as N-substituents — $CH_3NHCH_2CH_3$ is N-methylmethanamine… on the bigger chain: N-methylethanamine. The aromatic parent is **aniline** (benzenamine); ring-numbered derivatives: N,N-dimethylaniline.
- **Not to confuse:** amide ($-CONH_2$ — neutral acid derivative) vs amine ($-NH_2$ — basic). One letter changes the personality.

Preparation: Four Routes to the N–H

- **Ammonolysis of alkyl halides:** $R\text{–}X + NH_3 \rightarrow R\text{–}NH_2 + HX$ — but the fresh 1° amine attacks another halide, so the pot fills with 2°, 3° and quaternary salts. Useful for 1° with huge ammonia excess only; otherwise a mixture.
- **Reduction routes:** nitro compounds ($\xrightarrow{Sn/HCl}$, $\xrightarrow{Fe/HCl}$, or $\xrightarrow[Ni]{H_2}$) → 1° amines — **the** aniline route; nitriles and oximes with $LiAlH_4$; amides with $LiAlH_4$.
- **Gabriel phthalimide synthesis:** phthalimide's N–H (acidic) → potassium phthalimide → S_N2 with a primary halide → hydrolysis releases a **clean 1° aliphatic amine**. No 2°/3° contamination — but aryl halides refuse S_N2, so **aniline cannot be made this way**.
- **Hofmann bromamide reaction:** an amide + $\xrightarrow[KOH]{Br_2}$ → an amine with **one carbon fewer** — the carbonyl carbon leaves as carbonate: $RCONH_2 + Br_2 + 4KOH \rightarrow RNH_2 + K_2CO_3 + 2KBr + 2H_2O$. The migration step (R walks from C to N) makes it a named-reaction favourite.

Physical Properties: The H-Bond Ladder Again

- **Boiling points:** 1° and 2° amines hydrogen-bond (N–H donors), 3° amines cannot — so for the same mass, 1° > 2° > 3°. All sit below alcohols (N is less electronegative than O, weaker H-bonds).
- **Solubility:** the small amines mix with water (the lone pair accepts H-bonds); solubility sinks with chain length. Lower amines smell like rotten fish — literally ammonia's organic cousins.

Basic Character: The Twin Ladders

Setting	Order (most → least basic)	Why
Gas phase	$3° > 2° > 1° > NH_3$	pure inductive push — alkyl groups feed the lone pair
Aqueous (1°/2°/3° ethylamines)	$2° > 1° > 3° > NH_3$	solvation of the ion vs inductive push balance out; 3° loses (too crowded to hydrate)
Aryl vs alkyl	aliphatic >> aniline	aniline's lone pair sleeps in the ring by resonance
Ring-substituted anilines	p-toluidine > aniline > p-nitroaniline	donors raise, EWG lower; ortho effects complicate

- **The exam instruction:** BOTH ladders are correct — read the question's setting. "In gaseous phase" → the 3° ladder; "in aqueous solution" → the 2° ladder.
- **Practical consequence:** amines dissolve in dilute HCl as ammonium salts ($RNH_2 + HCl \rightarrow RNH_3^+Cl^-$) — the standard way to pull amines out of organic mixtures (and reverse with NaOH).

Chemical Properties of Amines

- **Salt formation:** with acids (see above); quaternary salts with excess alkyl halide ($R_3N + R'X \rightarrow R_3R'N^+X^-$) — long-chain quaternary salts work as detergents.
- **Acylation:** amine + acyl chloride/anhydride → **amide** (the lone pair attacks C=O). Friedel-Crafts cannot use aniline directly (it complexes with $AlCl_3$) — acylate nitrogen instead.
- **Carbylamine test (1° only):** amine + $CHCl_3$ + KOH (heat) → foul-smelling **isocyanide** (R–NC). 2° and 3° amines stay silent — a clean 1° detector (and a route to isocyanides).
- **Hinsberg test (benzenesulphonyl chloride):** 1° amine gives a product soluble in alkali; 2° gives an insoluble product; 3° reacts not at all — the classic three-way sorting hat.
- **Reaction with nitrous acid:** 1° **aliphatic** amines bubble out N_2 (alcohol forms) — a gas-evolution test; 1° **aromatic** amines (aniline, cold) build the **diazonium salt** instead — the gateway to the next section; 2° amines give yellow oily N-nitrosamines; 3° aromatic amines get nitrosated on the ring (p-nitroso).
- **Ring chemistry of aniline:** the —NH_2 group is a **hyper-activator** — bromine water instantly gives 2,4,6-tribromoaniline (white precipitate). To nitrate or sulphonate **selectively**, first **protect** the amine by acetylation (→ acetanilide), run the substitution, then hydrolyse back. Nitration without protection gives tar (oxidation) plus a meta mess; the protected route gives clean p-nitroaniline. Sulphonation gives sulphanilic acid — existing as a **zwitterion** ($H_3N^+–C_6H_4–SO_3^-$).

Diazonium Salts: The Aromatic Toolbox

Aliphatic diazonium salts explode even in solution — useless. Aromatic ones are stable cold, and they are **the** gateway for putting N, F, I, CN, OH or plain H exactly where an amino group once stood.



Figure 1: The diazonium factory: diazotisation conditions on the left; the seven synthesis exits on the right — Sandmeyer, Gattermann, Balz-Schiemann, KI, water, H_3PO_2, and azo coupling.

- **Diazotisation:** aniline + $NaNO_2$ + excess HCl at 273–278 K → benzenediazonium chloride. Cold, fresh, never isolated dry (explosive).
- **The seven exits (Figure 1):** Sandmeyer ($Cu_2\frac{Cl_2}{Cu_2}\frac{Br_2}{CuCN}$ → Cl, Br, CN); Gattermann (Cu powder, gentler); Balz-Schiemann (HBF_4, then heat → Ar–F); warm KI → Ar–I; warm water → phenol; H_3PO_2 or ethanol → benzene (H replaces the group — the eraser); coupling with phenol/aniline in mild base → **azo dyes** (p-hydroxyazobenzene — orange-red).
- **The conversion master-plan:** nitration (put NO_2) → reduction (put NH_2) → diazotisation → chosen exit. Need m-bromophenol? Direct methods fight the ring's preferences — the diazonium route just walks in.
- **Uses:** azo dyes (methyl orange), sulpha drugs (from sulphanilamide), quaternary ammonium detergents, and the biological amines — adrenaline, serotonin, histamine — that run your body.

Key Reactions & Rules: One-Stop Revision

PREPARATIONS, LADDERS, TESTS, EXITS

Rule / route	Symbols in plain words	When to reach for it
Ammonolysis: R–X + NH_3	excess NH_3 keeps 1°; else mixture	Alkyl amine synthesis
$\frac{Sn}{HCl}$ or $\frac{Fe}{HCl}$ on NO_2	nitro → amine	Aniline preparation

Rule / route	Symbols in plain words	When to reach for it
Gabriel phthalimide	clean 1° aliphatic only; aniline impossible	Pure 1° amine requests
Hofmann bromamide	amide + Br_2/KOH → amine, −1 carbon	Chain-shortening to amines
Gas basicity	3° > 2° > 1° > NH_3	"gaseous phase" questions
Aqueous basicity	2° > 1° > 3° > NH_3	"in water" questions
Aniline's weakness	lone pair delocalised into ring; aliphatic >> aniline	pKa comparisons
p-toluidine > aniline > p-nitroaniline	donors up, EWG down	Substituted aniline orders
Carbylamine: $CHCl_3$/KOH	1° only → foul R–NC	Detecting 1° amines
Hinsberg: $PhSO_2Cl$	1° soluble, 2° insoluble, 3° silent	Sorting the three classes
HNO_2 on amines	1° aliphatic: N_2 up; 1° aryl: diazonium; 2°: nitrosamine	Nitrous-acid test routes
Bromine water + aniline	2,4,6-tribromoaniline instantly	Hyper-activation check
Protect via acetylation	NH_2 → $NHCOCH_3$ before nitration	p-nitroaniline synthesis
Diazotisation	aniline + $NaNO_2$/HCl, 273–278 K	Every diazonium question
Sandmeyer: Cu_2X_2/CuCN	Ar–Cl, Ar–Br, Ar–CN	Halide and nitrile exits
Balz-Schiemann: HBF_4, heat	Ar–F	Fluoroarene synthesis
Warm H_2O / KI / H_3PO_2	phenol / Ar–I / benzene	OH, I, and erase exits
Coupling with phenol/aniline	azo dyes, orange-red	Dye synthesis questions

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
More alkyls = more basic, always	In WATER the order flips to $2° > 1° > 3°$	The setting (gas vs water) decides the ladder — read it first
Aniline is basic like other amines	Its lone pair sleeps in the ring — far weaker	pKa comparisons between aniline and ethylamine
Gabriel makes all amines	Fails for 2°/3° AND for aniline (aryl halides refuse S_N2)	"Gabriel for aniline" options are bait
Hofmann bromamide keeps the chain	It SHORTENS the chain by one carbon	Counting carbons in the product
All amines pass carbylamine	Only 1° amines do — 2°/3° are silent	Carbylamine as a 1° detector
Aniline nitrates cleanly like phenol	Unprotected aniline oxidises and meta-messes	The acetyl-protect-react-deprotect sequence
Diazonium salts keep like other salts	Aliphatic ones decompose; aromatic ones need cold and wet	"store dry" options are instant bait
F is impossible on rings directly	Balz-Schiemann does it via HBF_4	The fluoroarene synthesis route question

The Whole Chapter in One Picture

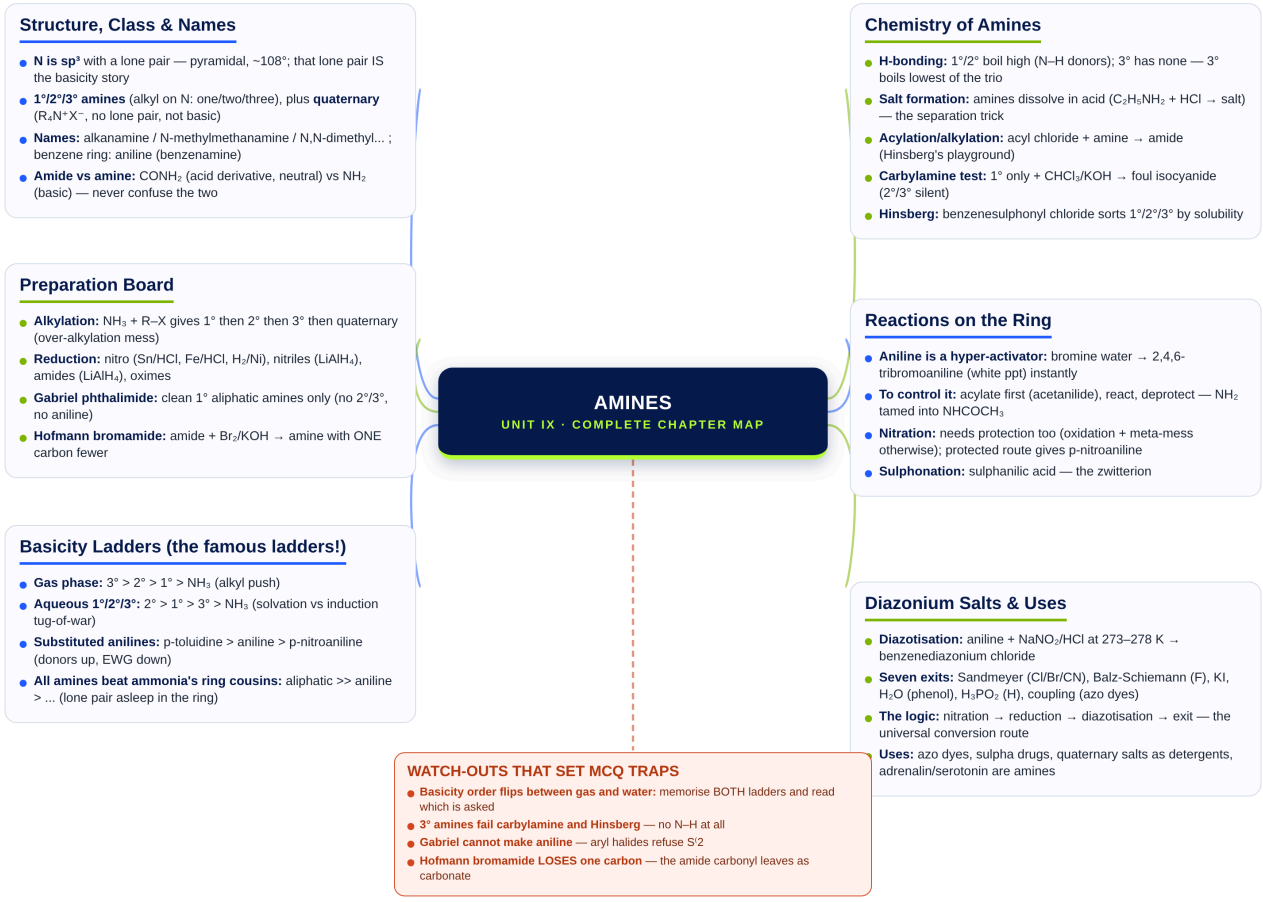


Figure 2: Chapter mindmap: structure and classification, the four preparation routes, the twin basicity ladders, amine tests and ring chemistry, diazonium exits, and uses — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. In the gaseous phase, the correct order of basicity is:

(A) $NH_3 > 1^\circ > 2^\circ > 3^\circ$
(B) $3^\circ > 2^\circ > 1^\circ > NH_3$
(C) $2^\circ > 1^\circ > 3^\circ > NH_3$
(D) $1^\circ > 2^\circ > 3^\circ > NH_3$

Q2. In aqueous solution, the basicity order of ethylamines is:

(A) $2^\circ > 1^\circ > 3^\circ > NH_3$
(B) $3^\circ > 2^\circ > 1^\circ > NH_3$
(C) $1^\circ > 2^\circ > 3^\circ > NH_3$
(D) $NH_3 > 1^\circ > 2^\circ > 3^\circ$

Q3. The carbylamine test is given by:

(A) primary amines only
(B) secondary amines only
(C) tertiary amines only
(D) all amines

Q4. Hofmann bromamide reaction on propanamide ($CH_3CH_2CONH_2$) gives:

(A) propanamine **(B)** ethanamine
(C) methanamine **(D)** N-methylmethanamine

Q5. Gabriel phthalimide synthesis CANNOT prepare:

(A) methylamine **(B)** ethylamine
(C) aniline **(D)** propylamine

Q6. Aniline is less basic than ethylamine because:

(A) aniline has a benzene ring that is heavy **(B)** the lone pair delocalises into the ring
(C) aniline cannot dissolve in water **(D)** ethylamine has more carbons

Q7. Bromine water added to aniline gives instantly:

(A) bromobenzene **(B)** 2,4,6-tribromoaniline
(C) p-bromoaniline only **(D)** no reaction without $FeBr_3$

Q8. The reagent pair for diazotisation of aniline is:

(A) $\frac{NaNO_2}{HCl}$ at 273–278 K **(B)** $\frac{NaNO_2}{HCl}$ at 373 K
(C) $\frac{HNO_3}{H_2SO_4}$ **(D)** $\frac{NH_3}{HCl}$

Q9. To put fluorine exactly where an NH_2 group stood, use:

(A) Sandmeyer with CuF **(B)** Balz-Schiemann: HBF_4, then heat
(C) direct fluorination **(D)** Swarts reaction

Q10. Coupling of benzenediazonium chloride with phenol in mild base gives:

(A) chlorobenzene **(B)** an azo dye (p-hydroxyazobenzene)
(C) phenyl cyanide **(D)** nitrobenzene

Detailed Solutions

Sol. Q1. Gas phase strips away solvation — only the inductive push remains, and three alkyl groups push hardest: $3° > 2° > 1° > NH_3$. The famous flip comes only when water enters.

Answer: (B)

Sol. Q2. In water, the ammonium ion must be stabilised by hydration. 2° balances push and solvation best; 3° is too crowded to hydrate and tumbles to third place: $2° > 1° > 3° > NH_3$.

Answer: (A)

Sol. Q3. Carbylamine needs the amine to supply TWO N–H hydrogens to build R–NC — only primary amines own them. It is the standard 1° detector (2°/3° stay silent).

Answer: (A)

Sol. Q4. Hofmann bromamide amputates the carbonyl carbon: the R group migrates to N. Propanamide → ethanamine, one carbon shorter — count before you answer.

Answer: (B)

Sol. Q5. Gabriel runs on S_N2 at an alkyl halide — aryl halides refuse it. Aniline is simply out of reach; aliphatic 1° amines like methylamine work cleanly.

Answer: (C)

Sol. Q6. Aniline's lone pair delocalises into the ring — partly asleep, less available for a proton. Ethylamine's pair sits fully on nitrogen. Delocalisation, not size, decides.

Answer: (B)

Sol. Q7. $-NH_2$ is a hyper-activator: bromine water reacts at all three o/p positions instantly, no catalyst — white 2,4,6-tribromoaniline precipitate. $FeBr_3$ is for plain benzene.

Answer: (B)

Sol. Q8. Diazotisation runs cold — 273–278 K with $NaNO_2$ in excess HCl. At higher temperatures the salt decomposes before you can use it.

Answer: (A)

Sol. Q9. Sandmeyer's copper salts deliver Cl, Br and CN only. Fluorine travels through Balz-Schiemann: build the diazonium tetrafluoroborate, then heat — N_2 leaves, F stays.

Answer: (B)

Sol. Q10. Coupling joins the diazonium cation to the activated ring of phenol (para position), forging the –N=N– bridge: p-hydroxyazobenzene, the orange-red azo dye.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 or Q2 mixed up	Using one ladder for both settings	Gas = 3° > 2° > 1°; water = 2° > 1° > 3° — the solvation tug-of-war flips it

If you answered...	Your slip was...	The corrected rule, in one line
Q4 option A	Forgetting the amputation	Hofmann bromamide always shortens the chain by exactly one carbon
Q6 option A	Judging by mass	Basicity lives in the lone pair's availability — ring delocalisation puts aniline's to sleep
Q7 option D	Applying benzene's rules	Aniline needs NO catalyst — it out-reacts even phenol with bromine water
Q9 option A	Stretching Sandmeyer too far	Copper delivers Cl/Br/CN only — fluorine always rides through HBF_4

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